EXHAUST SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5]

id0115q2800200

Warning

- **A hot engine and exhaust system can cause severe burns. Turn off the engine and wait until they are cool before removing the exhaust system.**

Caution

- **Perform exhaust gas temperature sensor removal/installation only when inspection or replacement is required, such as when DTCs for the sensor are displayed, because it may cause exhaust gas leakage. If the exhaust gas temperature sensor is removed/installed once, always verify that there is no exhaust gas leakage after reinstallation.**

Note

- If the catalytic converter (DPF) is replaced, perform the “Catalytic Converter (DPF) Replacement Procedure”.
- If the exhaust shutter valve is replaced, perform the “Exhaust Shutter Valve Replacement Procedure”.

Catalytic Converter (DPF) Replacement Procedure

1. If the catalytic converter (DPF) is replaced, perform the following procedure.

STEP	ACTION	PAGE/CONDITION
1	Perform DPF data reset procedure.	(See DIESEL PARTICULATE FILTER DATA RESET [SKYACTIV-D 1.5].)
2	Switch the ignition off.	—
3	Wait for 20 s or more.	—
4	Perform KOEO self-test procedure.	(See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].)
5	Perform KOER self-test procedure.	(See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].)
6	Perform fuel injector injection amount correction.	(See FUEL INJECTOR INJECTION AMOUNT CORRECTION [SKYACTIV-D 1.5].)
7	Perform compulsory diesel particulate filter regeneration.	(See COMPULSORY DIESEL PARTICULATE FILTER REGENERATION [SKYACTIV-D 1.5].)
8	Clear the DTCs.	(See OBD READ/CLEAR DIAGNOSTIC TEST RESULTS [SKYACTIV-D 1.5].)

Exhaust Shutter Valve Replacement Procedure

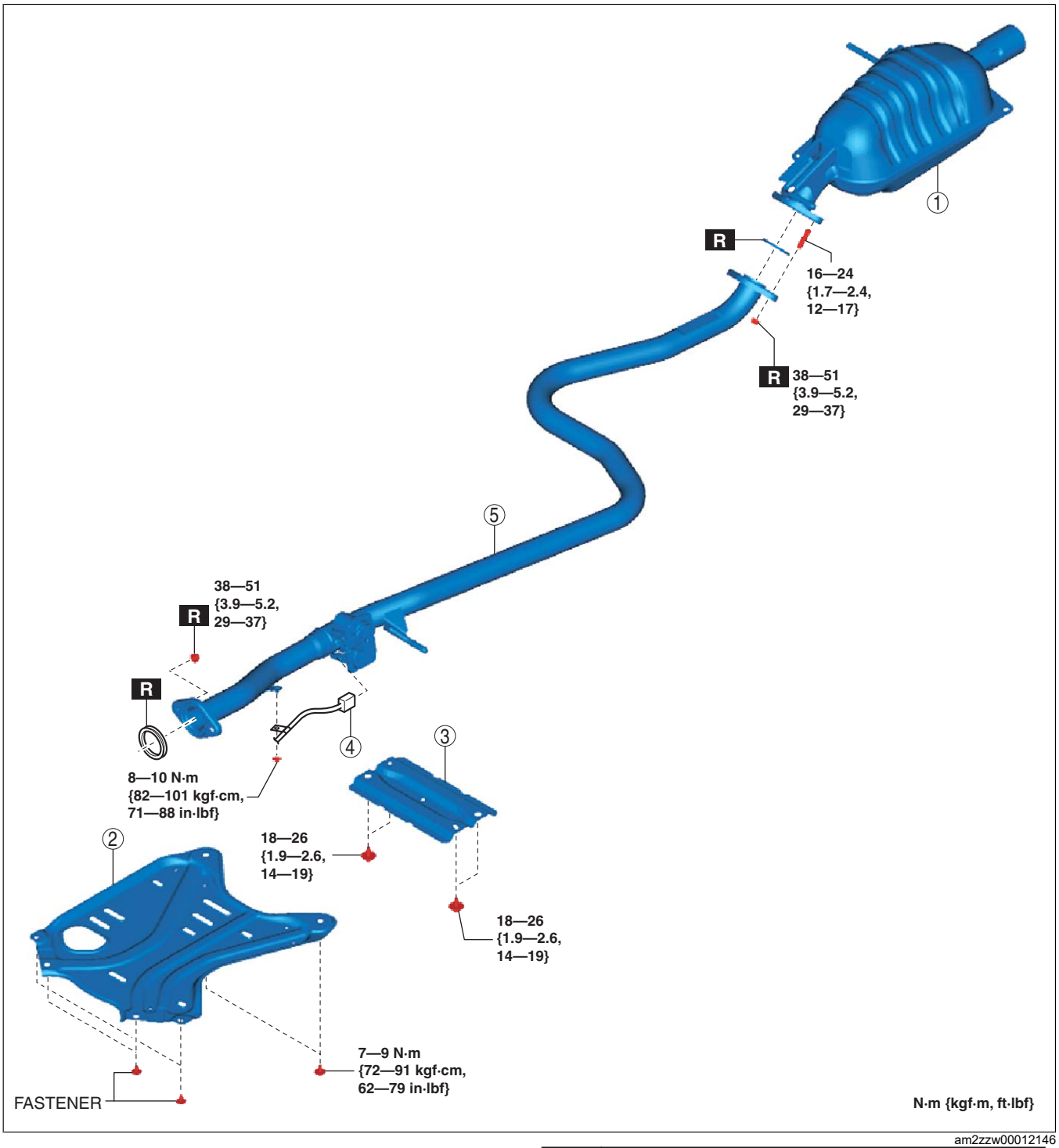
1. If the exhaust shutter valve is replaced, perform the following procedure.

STEP	ACTION	PAGE/CONDITION
1	Perform exhaust shutter valve initialization procedure.	(See EXHAUST SHUTTER VALVE INITIALIZATION [SKYACTIV-D 1.5].)
2	Warm the engine until the engine coolant temperature is 70 degrees C.	—
3	Switch the ignition off.	—
4	Wait for 20 s or more.	—
5	Start the engine.	—
6	Maintain the engine idling condition for 10 s.	—
7	Switch the ignition off.	—
8	Wait for 20 s or more.	—
9	Perform KOEO self-test procedure.	(See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].)
10	Perform KOER self-test procedure.	(See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].)

Exhaust System Removal /Installation

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/ CONNECTION.)
2. Drain the engine coolant. (See ENGINE COOLANT REPLACEMENT [SKYACTIV-D 1.5].)
3. Remove the front under cover No.2. (See FRONT UNDER COVER No.2 REMOVAL/INSTALLATION.)
4. Remove in the order shown in the figure.
5. Remove the exhaust system insulator. (See Exhaust system insulator removal/installation.)
6. Install in the reverse order of removal.

Step 1

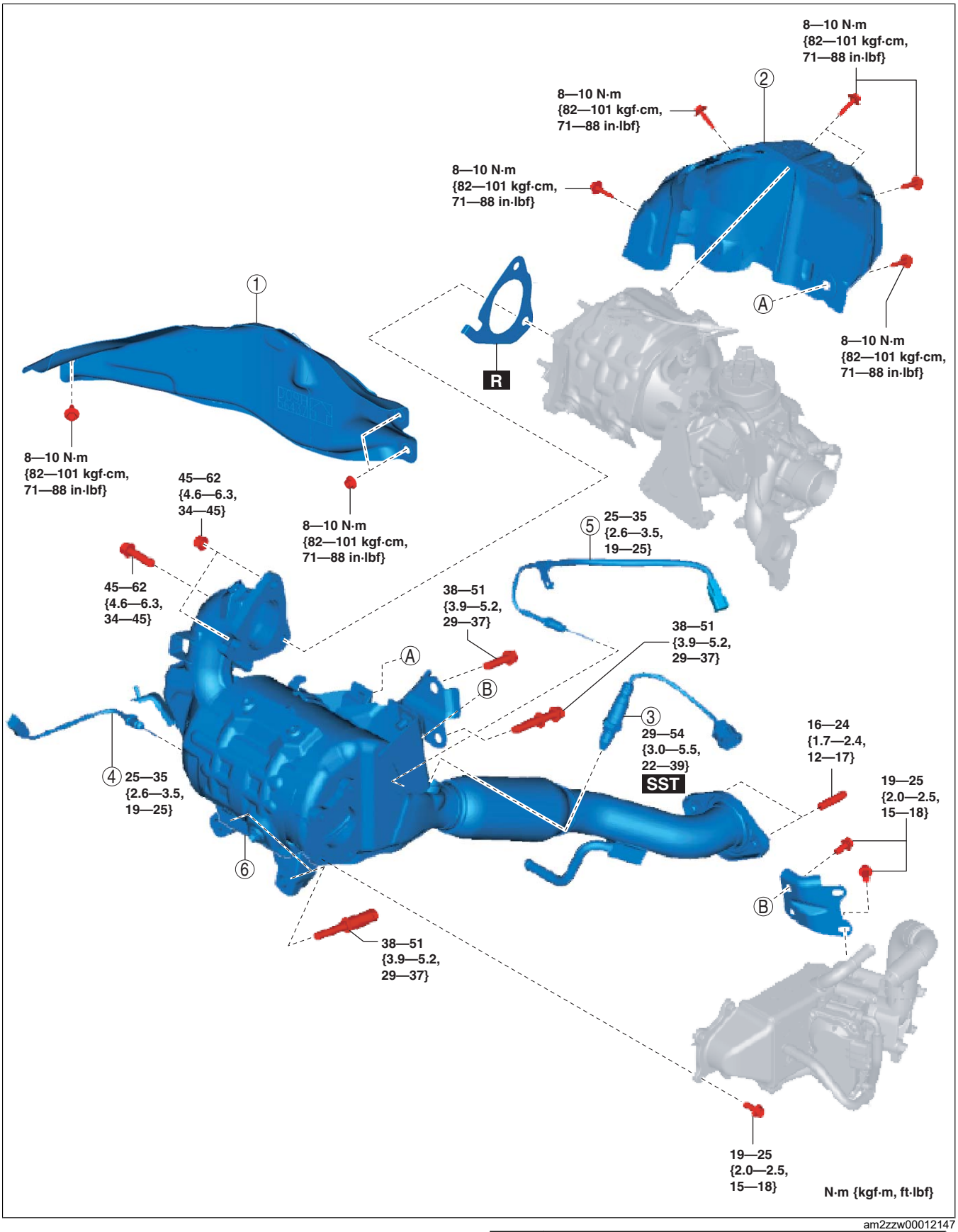


am2zzw00012146

1	Main silencer
2	Plate
3	Tunnel member

4	Exhaust shutter valve
5	Middle pipe

Step 2



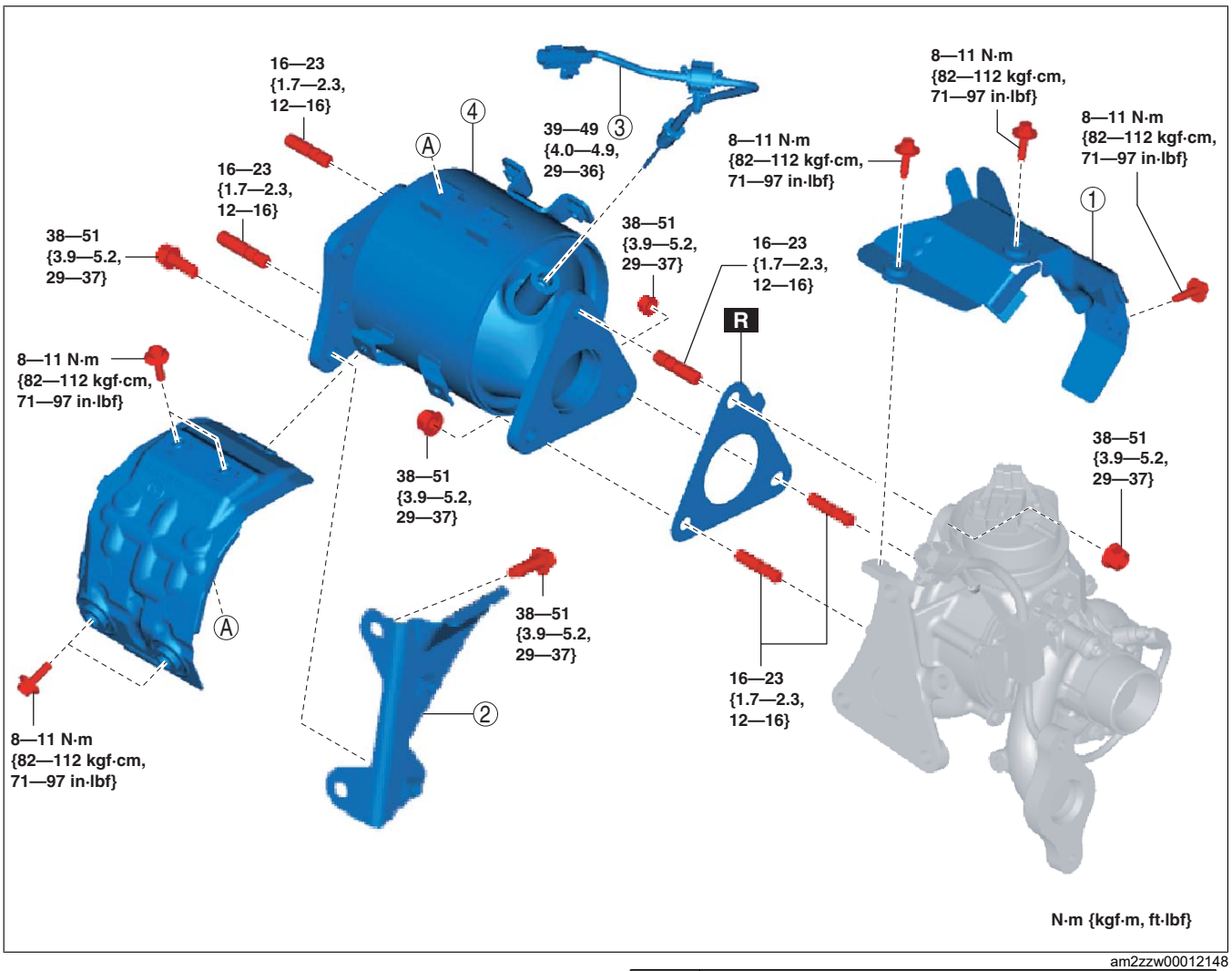
am2zzw00012147

1	Insulator

2	Catalytic converter (DPF) insulator (See Catalytic converter (DPF) Insulator removal note.) (See Catalytic converter (DPF) insulator installation note.)

3	A/F sensor (See AIR FUEL RATIO (A/F) SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
4	Exhaust gas temperature sensor No.3 (See EXHAUST GAS TEMPERATURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
5	Exhaust gas temperature sensor No.4 (See EXHAUST GAS TEMPERATURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
6	Catalytic converter (DPF) (See Catalytic converter (DPF) removal note.) (See Catalytic converter (DPF) installation note.) (See Catalytic Converter (DPF) Replacement Procedure.)

Step 3

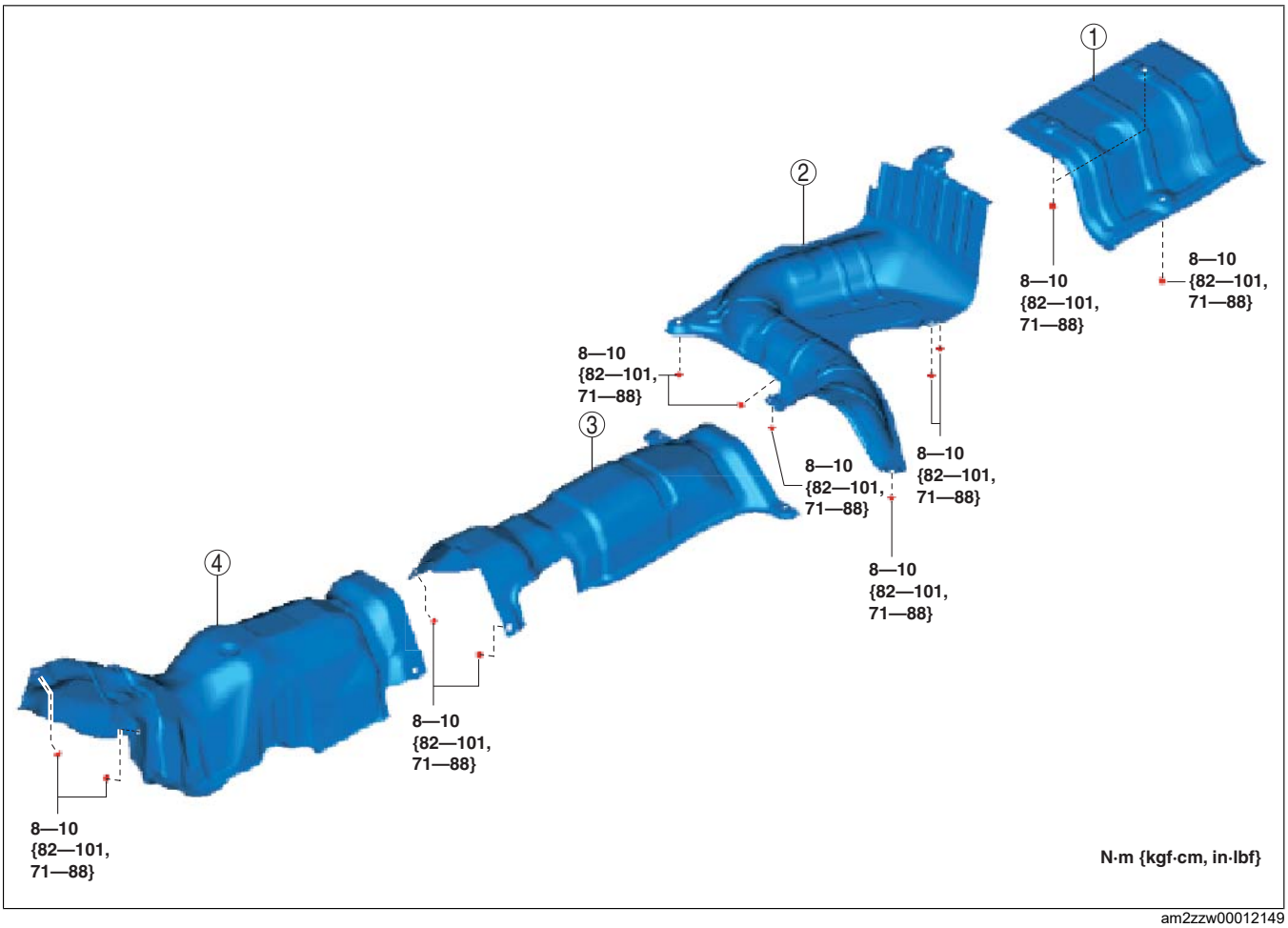


am2zzw00012148

1	Turbocharger insulator
2	Catalytic converter (oxidation catalyst) bracket (See Catalytic converter (oxidation catalyst) bracket removal note.) (See Catalytic converter (oxidation catalyst) bracket installation note.)
3	Exhaust gas temperature sensor No.2 (See EXHAUST GAS TEMPERATURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
4	Catalytic converter (oxidation catalyst) (See Catalytic converter (oxidation catalyst) installation note.)

Exhaust system insulator removal/installation

1. Remove in the order shown in the figure.
2. Install in the reverse order of removal.



am2zzw00012149

1	Insulator (rear)
2	Insulator (middle No.1)
3	Insulator (middle No.2)
4	Insulator (front)

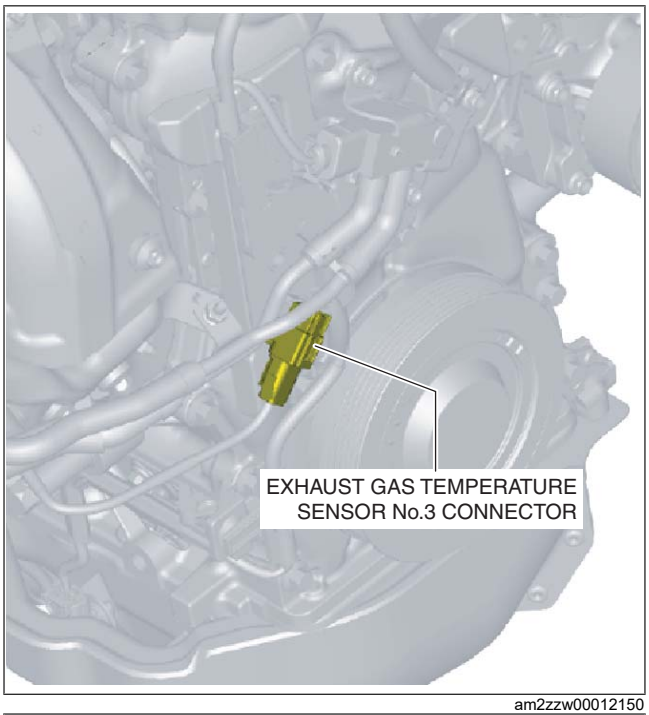
Catalytic converter (DPF) Insulator removal note

1. Remove the windshield wiper arm and blade. (See WINDSHIELD WIPER ARM AND BLADE REMOVAL/INSTALLATION.)
2. Remove the cowl grille. (See COWL GRILLE REMOVAL/INSTALLATION.)
3. Remove the windshield wiper motor and link. (See WINDSHIELD WIPER MOTOR AND LINK REMOVAL/INSTALLATION.)
4. Remove the cowl panel. (See COWL PANEL REMOVAL/INSTALLATION.)
5. Set the PCM aside so that it does not interfere with the servicing. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
6. Set the cooler hose (LO) aside so that it does not interfere with the servicing. (See REFRIGERANT LINE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
7. Remove the catalytic converter (DPF) insulator.

Catalytic converter (DPF) removal note

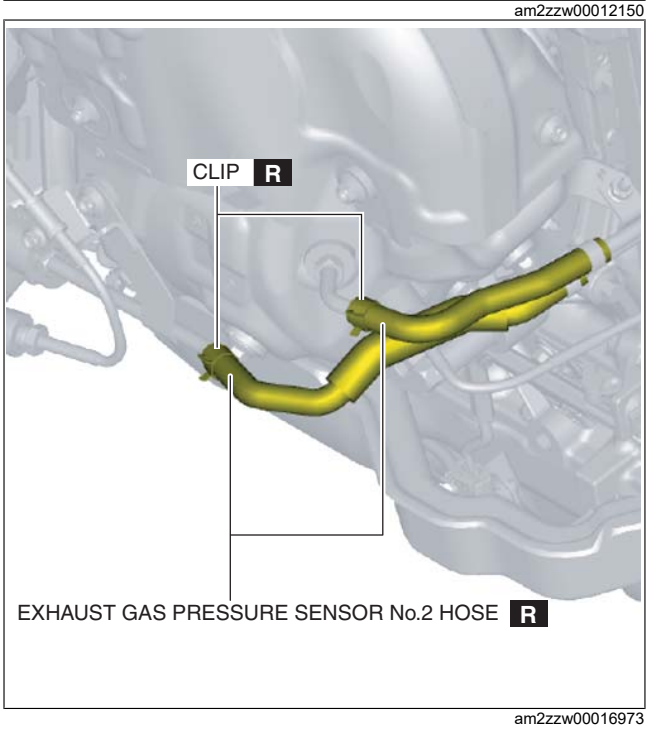
1. Remove the front crossmember. (See FRONT CROSSMEMBER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
2. Remove the battery and the battery tray. (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
3. Set the DC-DC converter (i-ELOOP) aside so that it does not interfere with the servicing. (With i-ELOOP) (See DC-DC CONVERTER (i-ELOOP) REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].)
4. Disconnect the LP-EGR pipe from the turbocharger air inlet hose. (See LP-EGR CONTROL VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
5. Remove the drive shaft bracket. (See FRONT DRIVE SHAFT REMOVAL/INSTALLATION.)

6. Disconnect exhaust gas temperature sensor No.3 connector shown in the figure.(See EXHAUST GAS TEMPERATURE SENSOR REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)



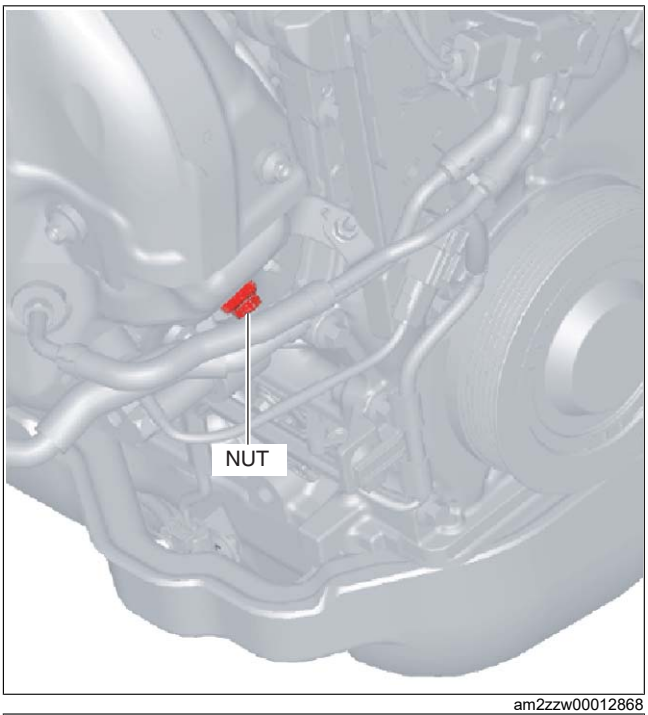
am2zzw00012150

7. Disconnect the exhaust gas pressure sensor No.2 hose on the catalytic converter (DPF) side shown in the figure.(See EXHAUST GAS TEMPERATURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)



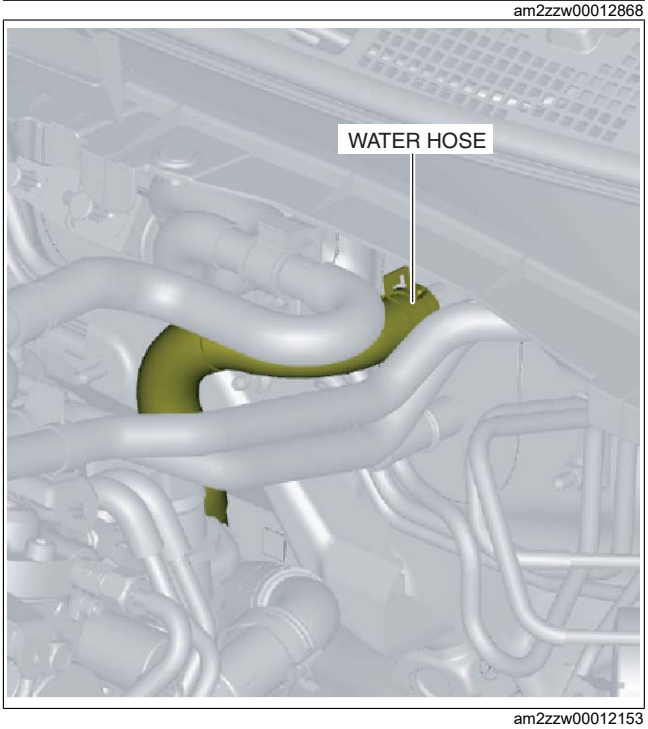
am2zzw00016973

8. Remove the nut shown in the figure and set the bracket aside.
9. Disconnect the connectors of the following parts:
   - LP-EGR control valve (See LP-EGR CONTROL VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
   - A/F sensor (See AIR FUEL RATIO (A/F) SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
   - Exhaust gas temperature No.4 (See EXHAUST GAS TEMPERATURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)



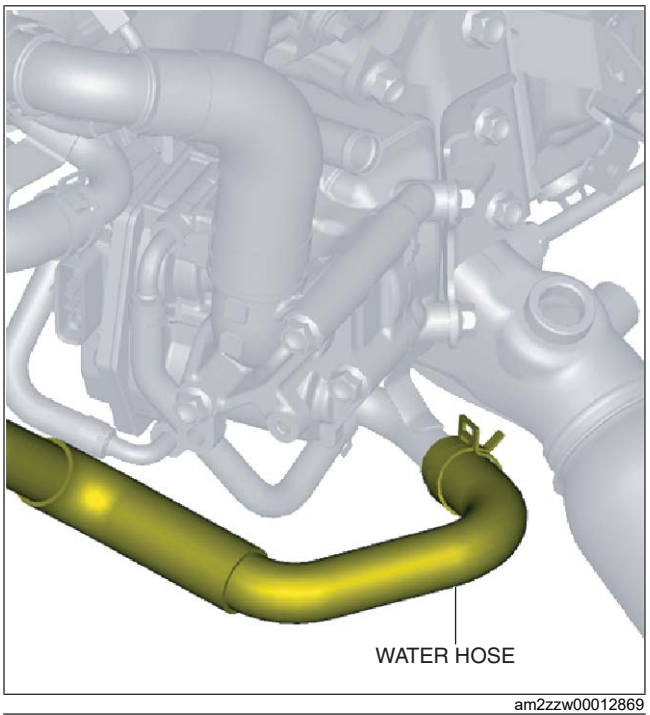
am2zzw00012868

10. Disconnect the water hose shown in the figure.



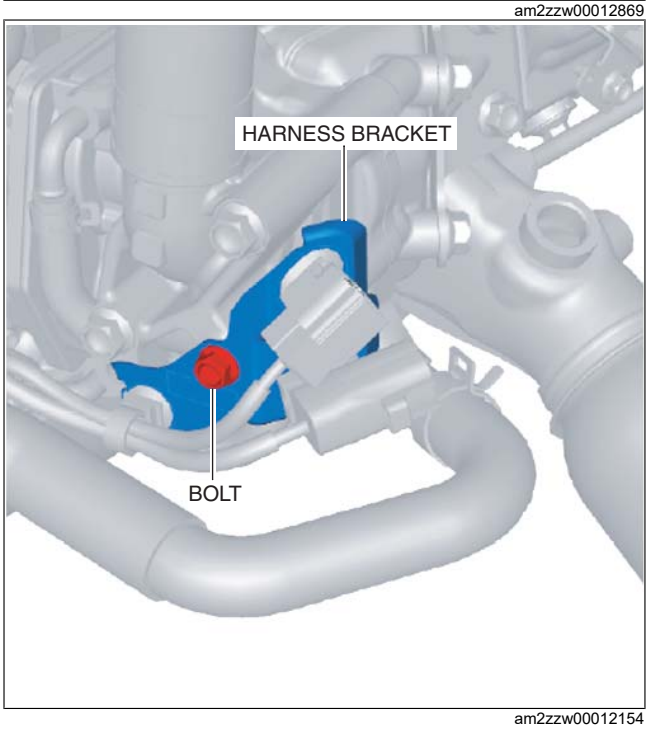
am2zzw00012153

11. Disconnect the water hose shown in the figure.



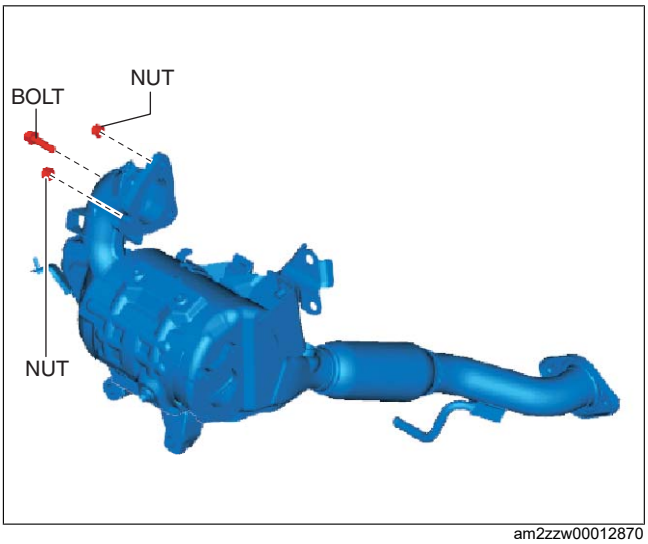
am2zzw00012869

12. Remove the bolts shown in the figure and set the wiring harness bracket aside.



am2zzw00012154

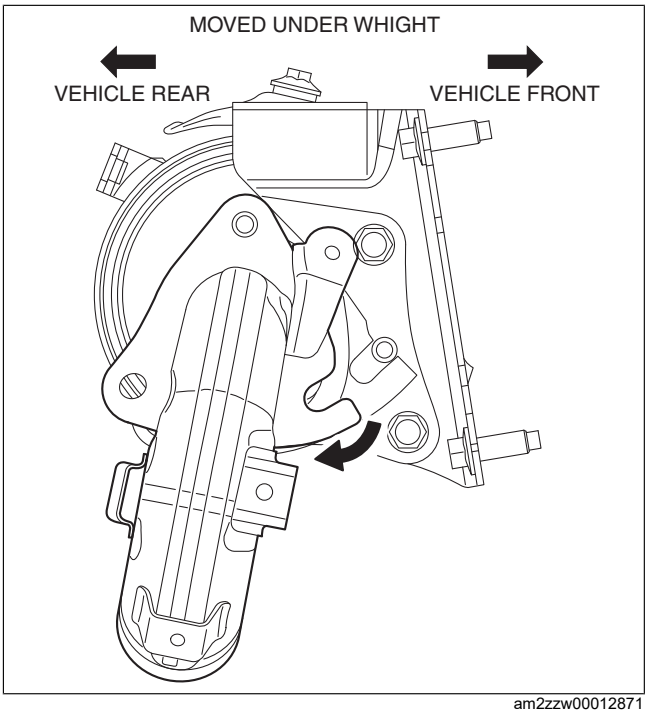
13. Remove the catalytic converter (DPF) installation nuts and bolt.



am2zzw00012870

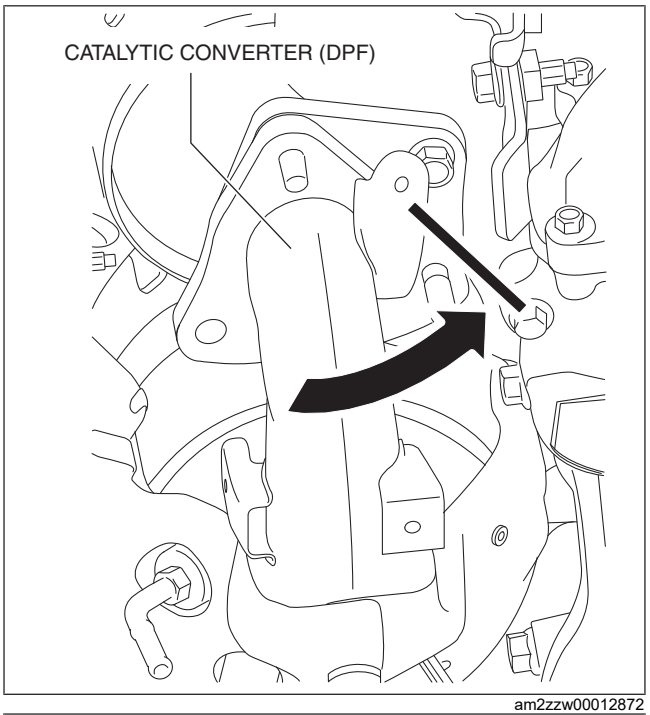
Note

- When the nuts and bolts of the catalytic converter (DPF) are removed, the part will move under its own weight as shown in the figure.



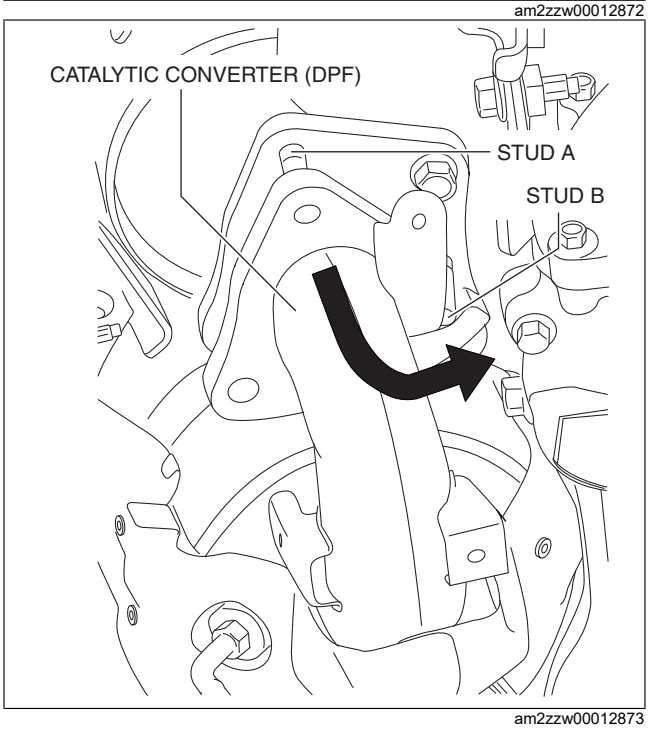
am2zzw00012871

14. Move the catalytic converter (DPF) to the position shown in the figure.



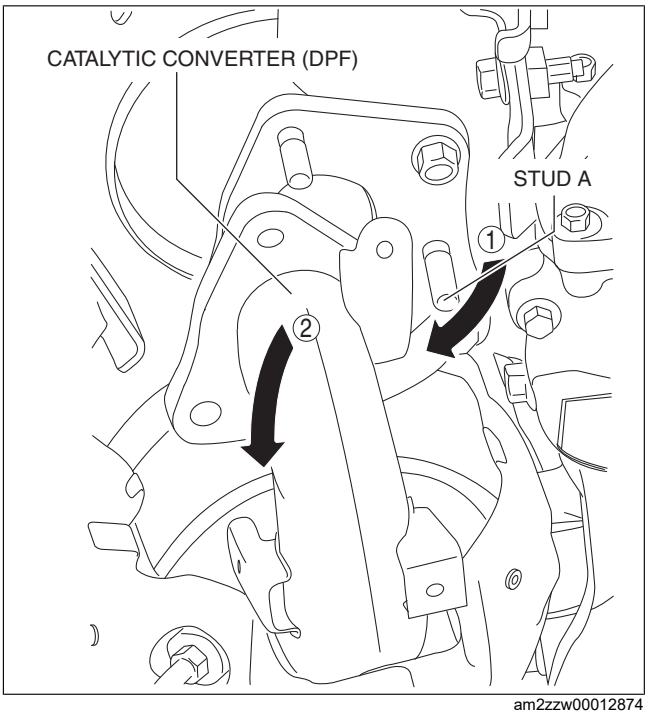
am2zzw00012872

15. Pivoting on stud A, move the catalytic converter (DPF) in the direction of the arrow shown in the figure until stud B is removed.



am2zzw00012873

16. Move the catalytic converter (DPF) in the direction of arrow 1 along the notch groove adjoining stud A, and remove the catalytic converter (DPF) in the direction of arrow 2 as shown in the figure.



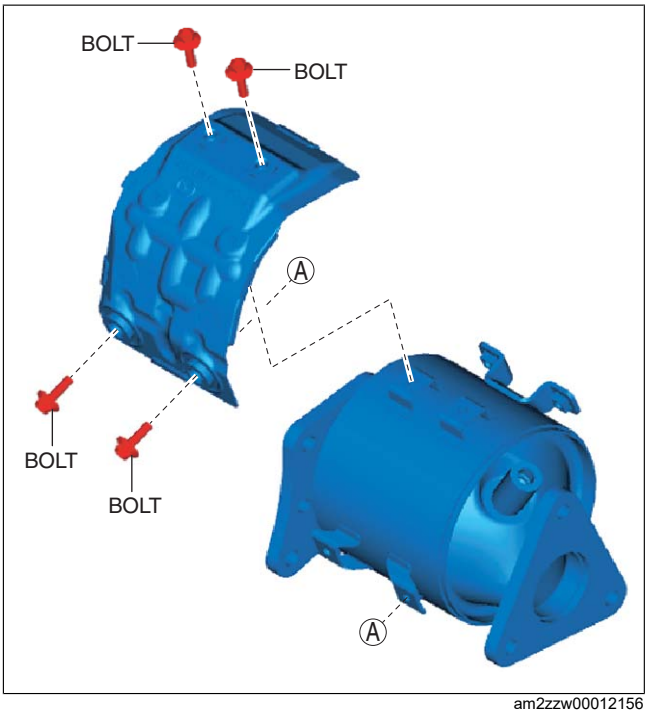
am2zzw00012874

Catalytic converter (oxidation catalyst) bracket removal note

1. Remove the nuts and bolts of the catalytic converter (oxidation catalyst).
2. Remove the catalytic converter (oxidation catalyst) bracket.

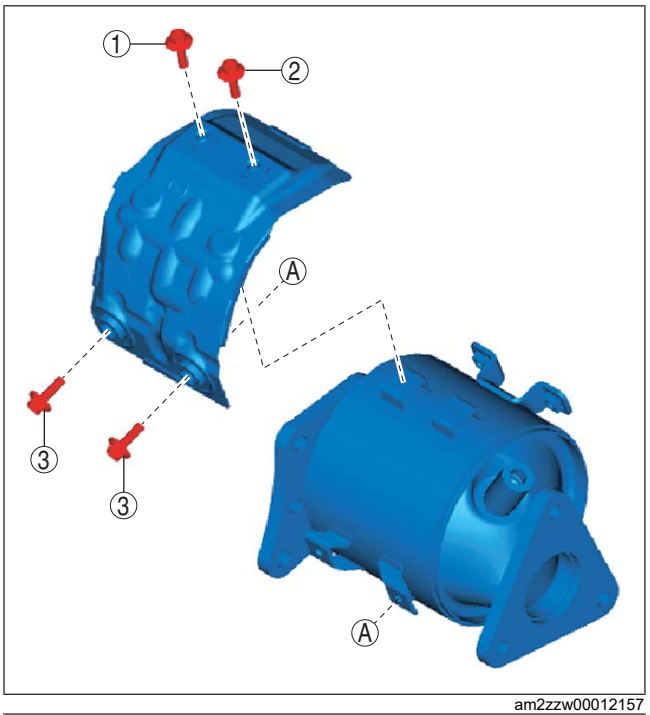
Catalytic converter (oxidation catalyst) installation note

1. Temporarily tighten the bolts shown in the figure until they contact the opposing part.



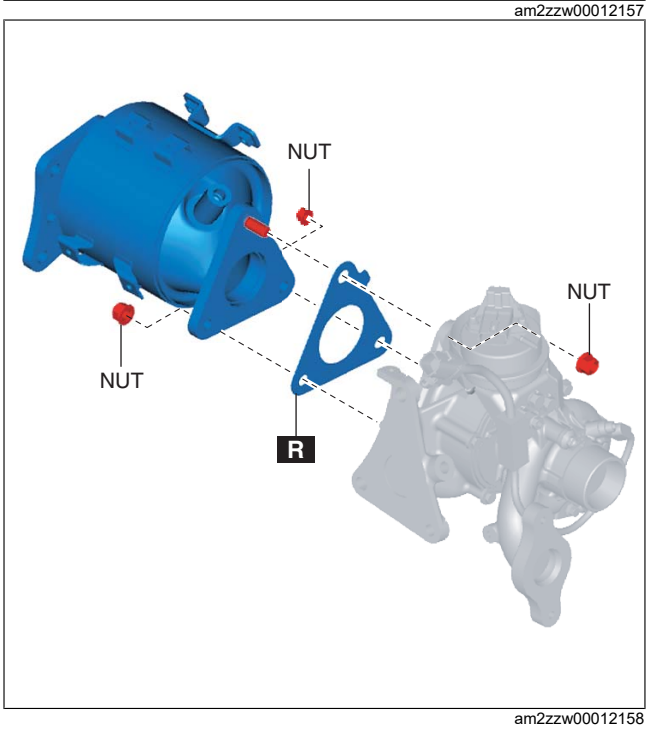
am2zzw00012156

2. Temporarily tighten the bolts in the order shown in the figure.



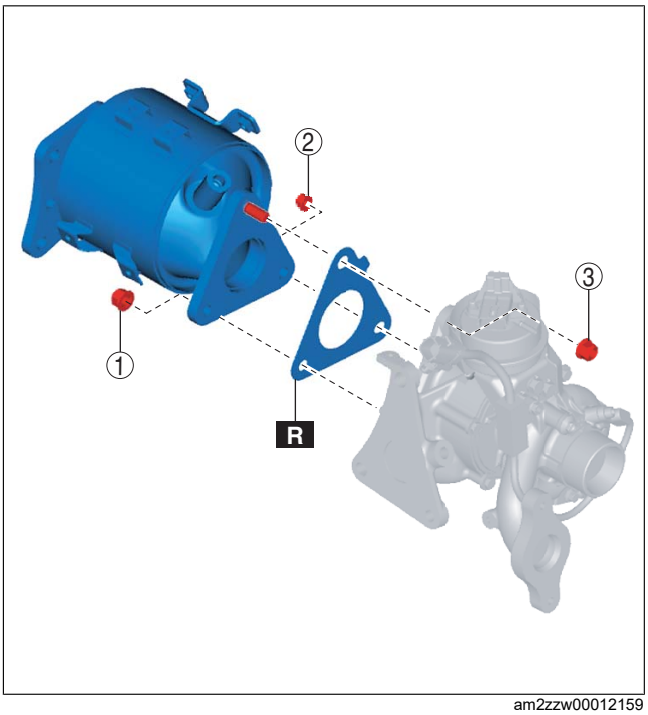
am2zzw00012157

3. Temporarily tighten the nuts shown in the figure until they contact the opposing part.



am2zzw00012158

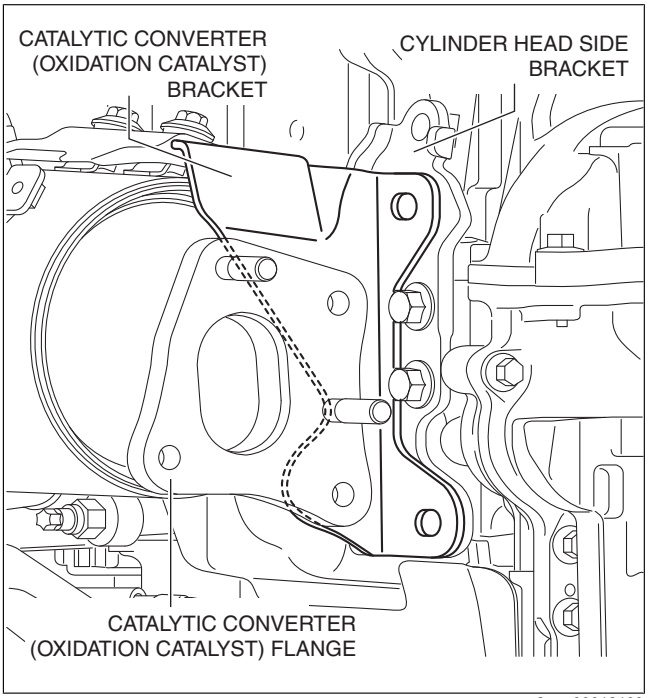
4. Install the nuts in the order shown in the figure.



am2zzw00012159

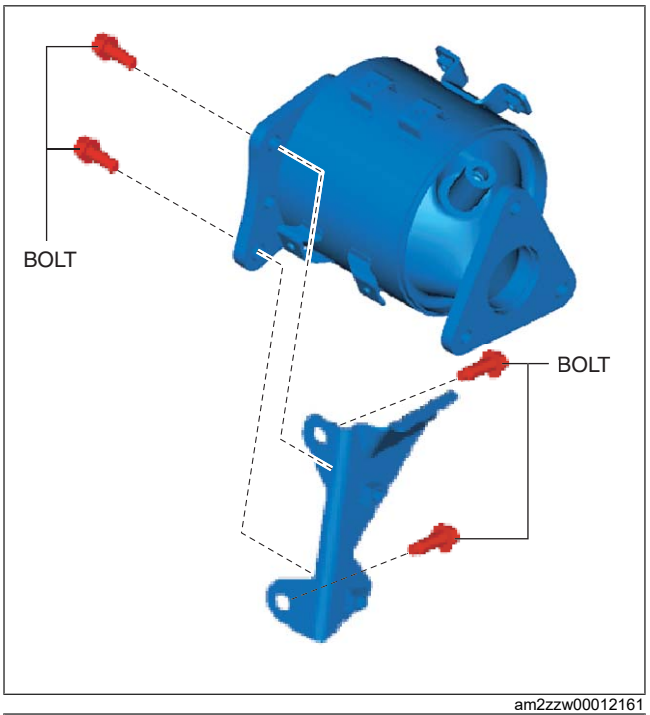
Catalytic converter (oxidation catalyst) bracket installation note

1. Align the catalytic converter (oxidation catalyst) bracket along the flange and cylinder head side bracket of the catalytic converter (oxidation catalyst) position.



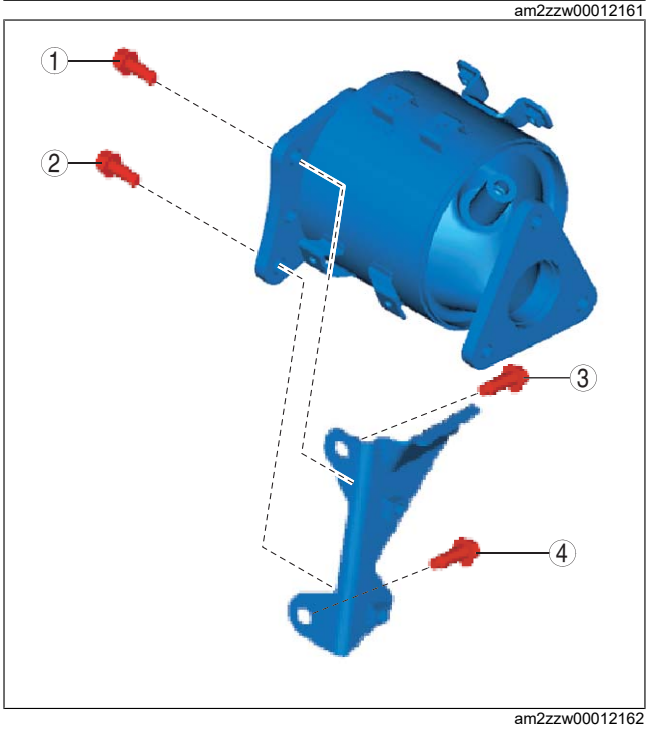
am2zzw00012160

2. Temporarily tighten the bolts shown in the figure until they contact the opposing part.



am2zzw00012161

3. Temporarily tighten the bolts in the order shown in the figure.



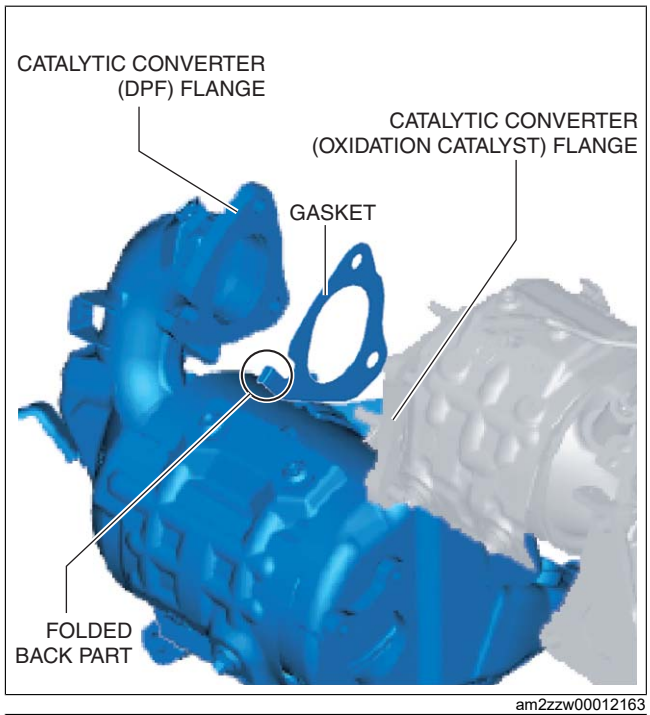
am2zzw00012162

Catalytic converter (DPF) installation note

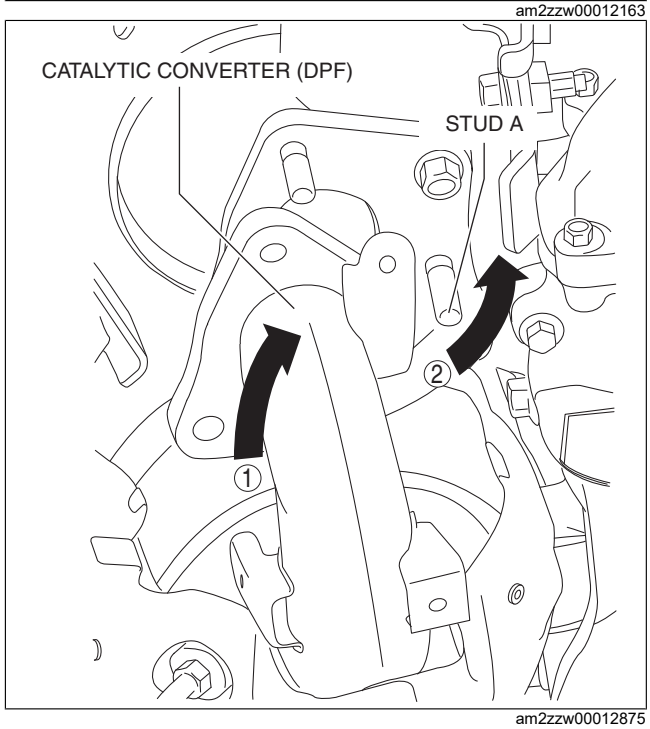
Note

- Do not let the folded back part of the gasket become pinched in the catalytic converter (DPF) flange and catalytic converter (oxidation catalytic converter) flange

1. Move the catalytic converter (DPF) in the direction of arrow 2 along the notch groove adjoining stud A, and install the catalytic converter (DPF) in the direction of arrow 1 as shown in the figure.

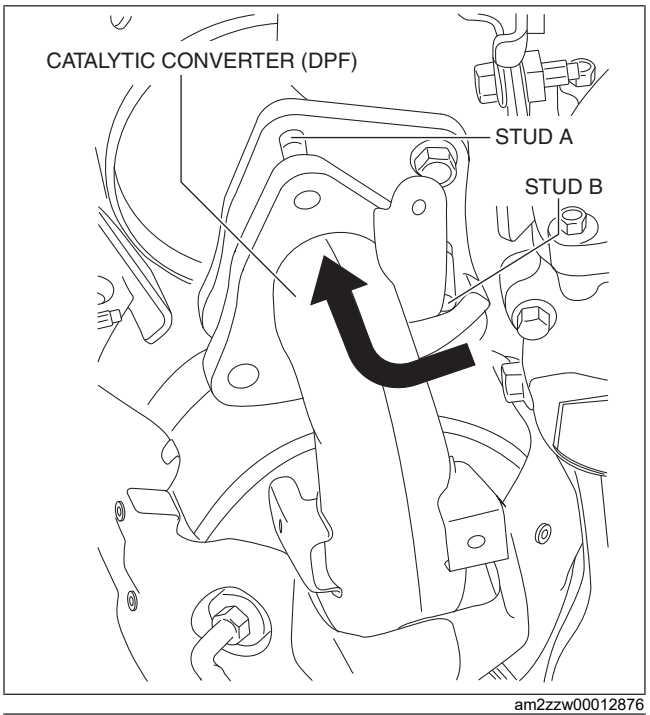


am2zzw00012163



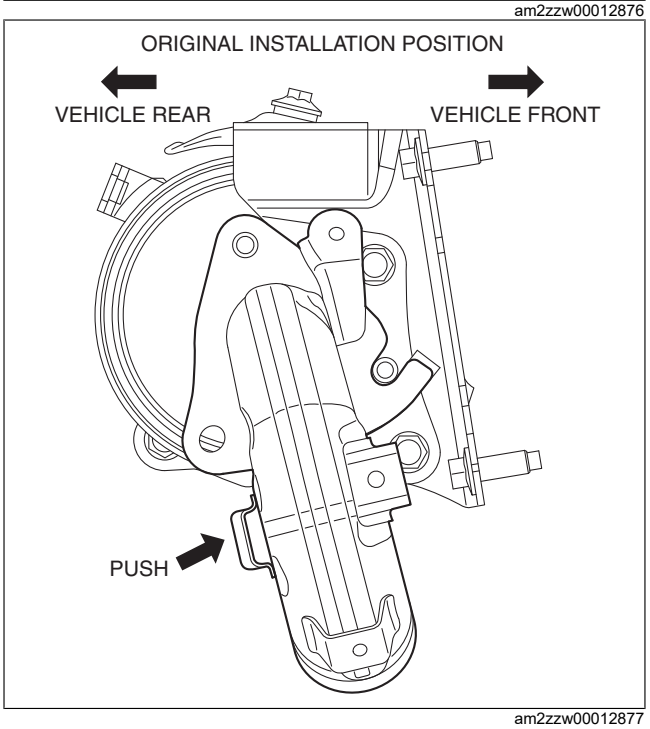
am2zzw00012875

2. Install the catalytic converter (DPF) to stud B by pivoting it in the direction of the arrow on stud A.



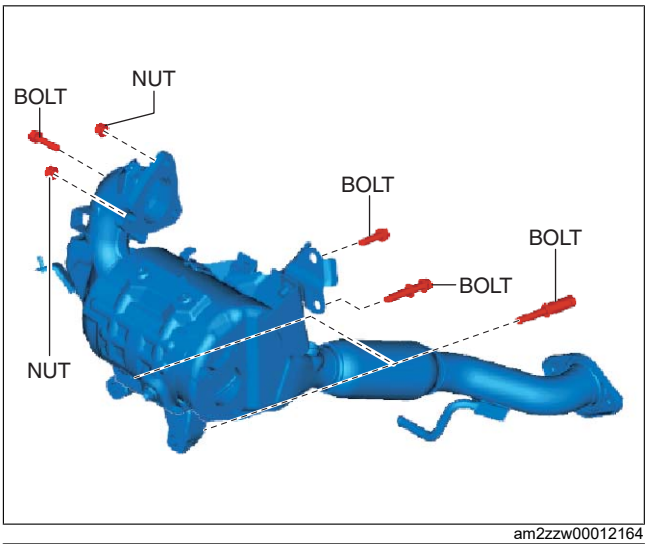
am2zzw00012876

3. Move the catalytic converter from the position where it rests under its own weight to its original installation position.



am2zzw00012877

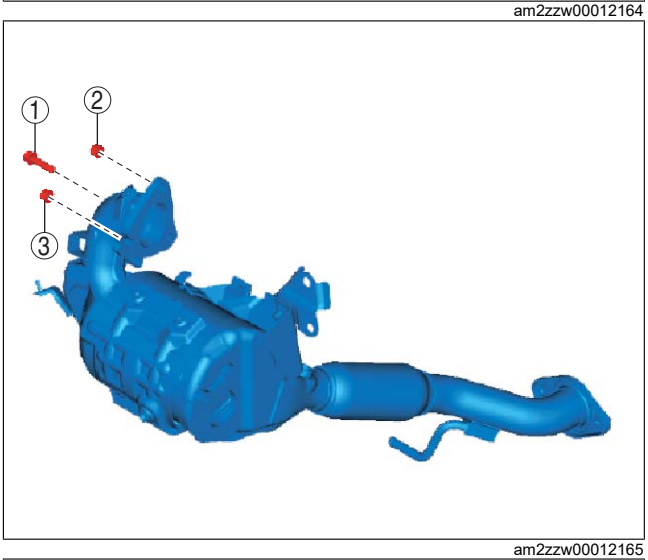
4. Temporarily tighten the bolts and nuts shown in the figure until they contact the opposing part.



am2zzw00012164

5. Install the bolts and nuts in the order shown in the figure.

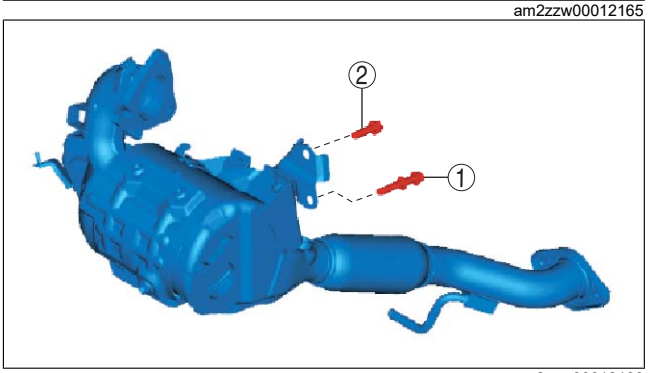
Tightening torque
45—62 N·m {4.6—6.3 kgf·m, 34—45 ft·lbf}



am2zzw00012165

6. Temporarily tighten the bolts in the order shown in the figure.

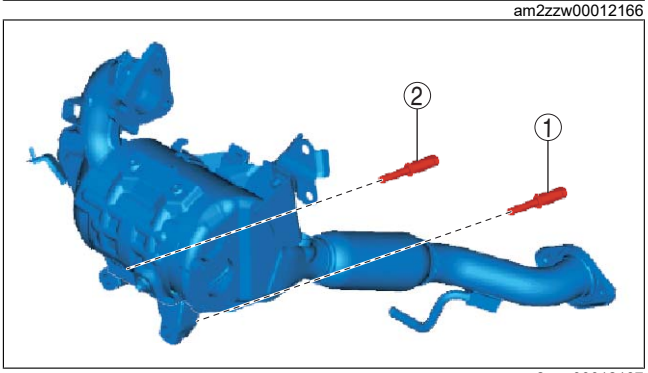
Tightening torque
38—51 N·m {3.9—5.2 kgf·m, 29—37 ft·lbf}



am2zzw00012166

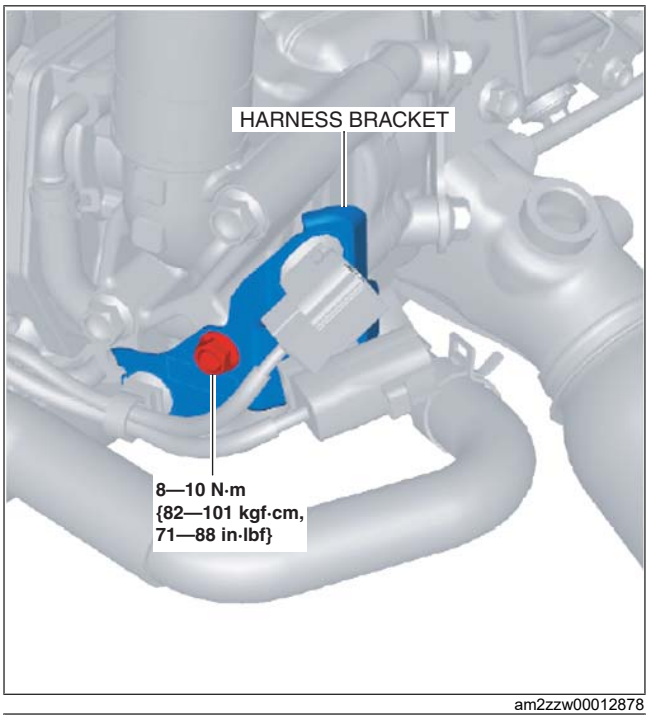
7. Temporarily tighten the bolts in the order shown in the figure.

Tightening torque
38—51 N·m {3.9—5.2 kgf·m, 29—37 ft·lbf}



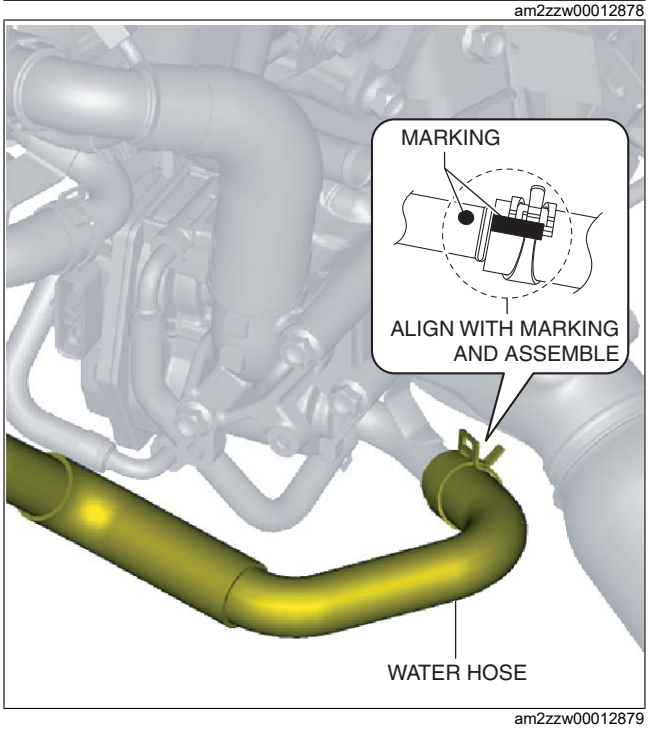
am2zzw00012167

8. Install the harness bracket shown in the figure.



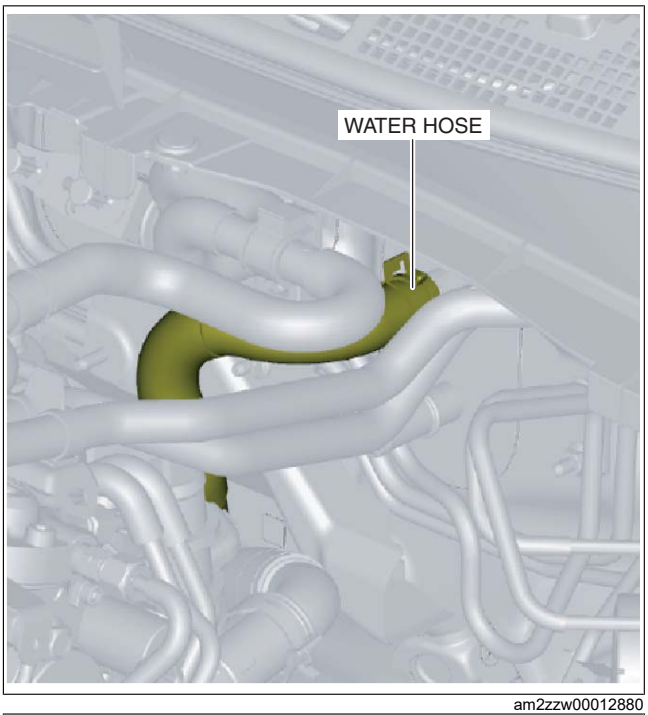
am2zzw00012878

9. Install the water hose shown in the figure.



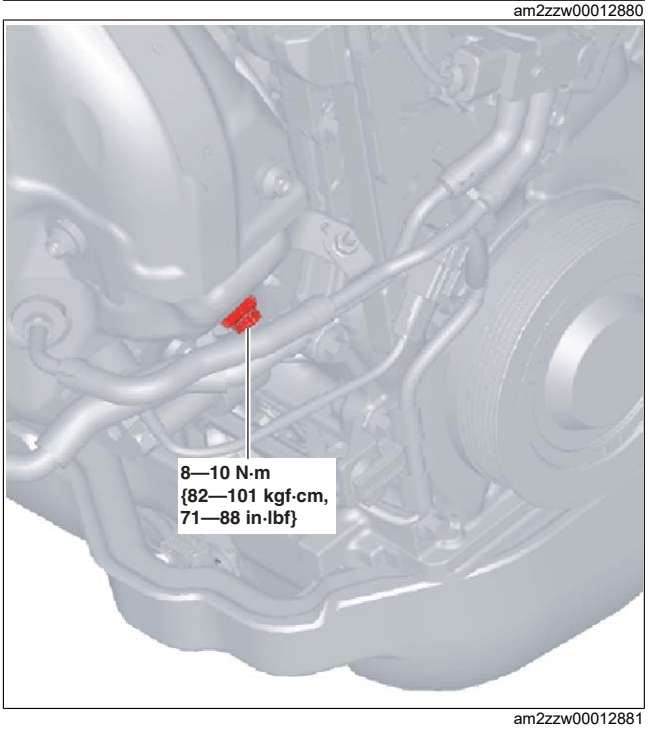
am2zzw00012879

10. Install the water hose shown in the figure.
11. Connect the connectors of the following parts:
    - Exhaust gas temperature No.4 (See EXHAUST GAS TEMPERATURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
    - A/F sensor (See AIR FUEL RATIO (A/F) SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
    - LP-EGR control valve (See LP-EGR CONTROL VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)



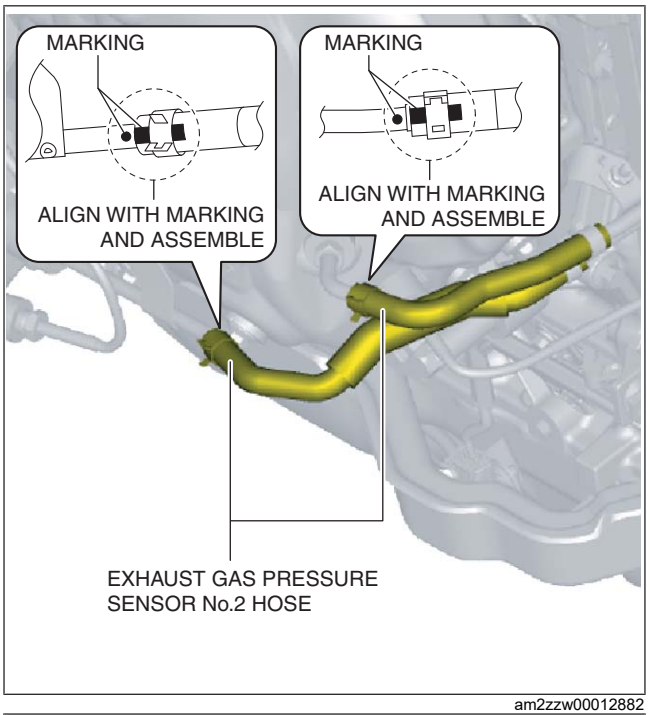
am2zzw00012880

12. Install the nut shown in the figure.



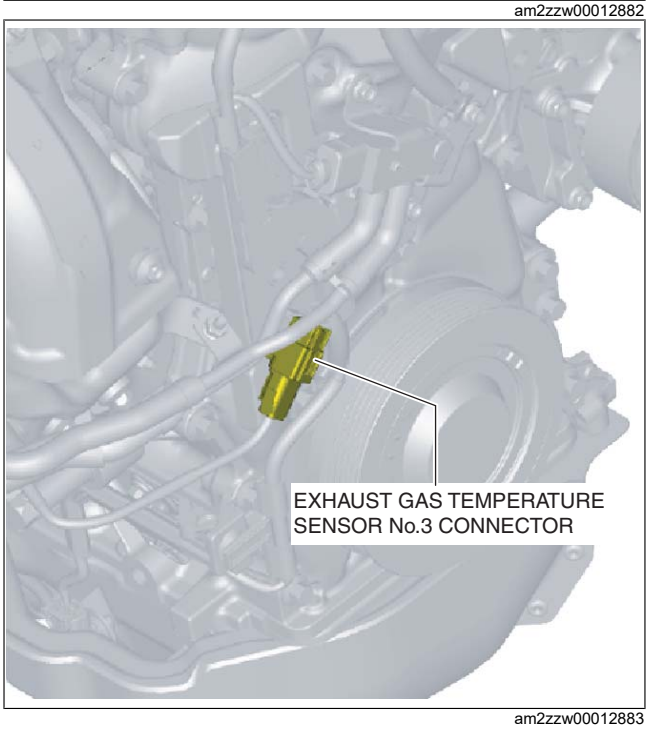
am2zzw00012881

13. Connect the hose of the exhaust gas pressure sensor No.2 shown in the figure.



am2zzw00012882

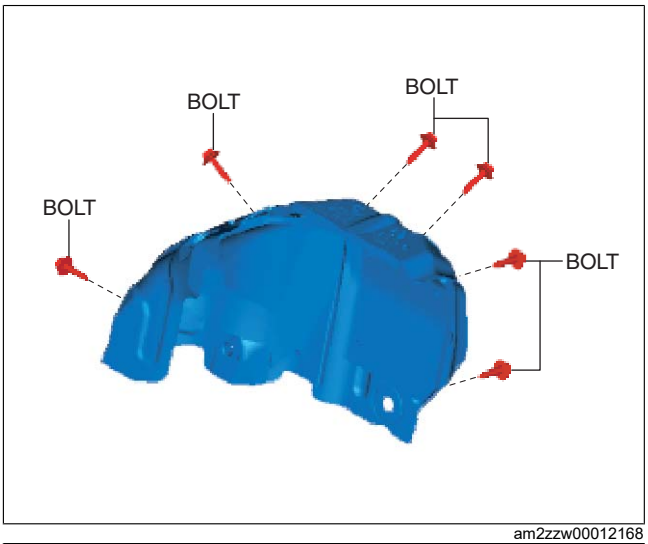
14. Connect the hose of the exhaust gas temperature sensor No.3 connector shown in the figure.
15. Install the drive shaft bracket. (See FRONT DRIVE SHAFT REMOVAL/INSTALLATION.)
16. Install the LP-EGR pipe. (See LP-EGR CONTROL VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
17. Install the DC-DC converter (i-ELOOP). (With i-ELOOP) (See DC-DC CONVERTER (i-ELOOP) REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].)
18. Install the battery and the battery tray. (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
19. Install the front crossmember. (See FRONT CROSSMEMBER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)



am2zzw00012883

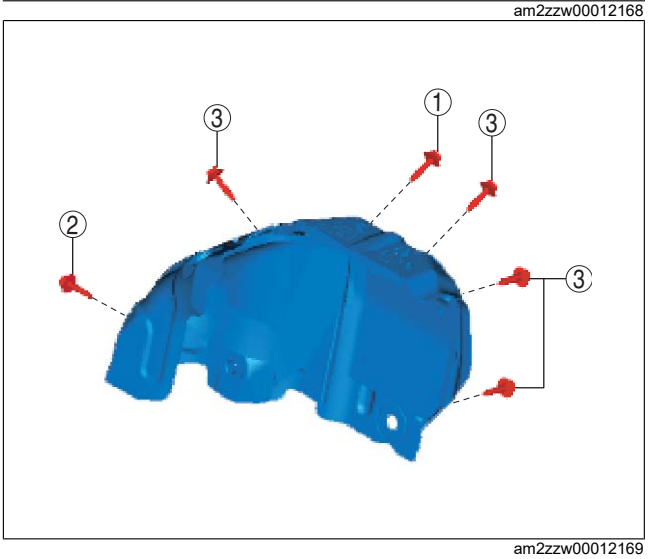
Catalytic converter (DPF) insulator installation note

1. Temporarily tighten the bolts shown in the figure until they contact the opposing part.



am2zzw00012168

2. Temporarily tighten the bolts in the order shown in the figure.



am2zzw00012169

Tightening torque
8—10 N·m {82—101 kgf·cm, 71—88 in·lbf}

3. Install the cooler hose (LO). (See REFRIGERANT LINE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
4. Install the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
5. Install the cowl panel. (See COWL PANEL REMOVAL/INSTALLATION.)
6. Install the windshield wiper motor and link. (See WINDSHIELD WIPER MOTOR AND LINK REMOVAL/INSTALLATION.)
7. Install the cowl grille. (See COWL GRILLE REMOVAL/INSTALLATION.)
8. Install the windshield wiper arm and blade. (See WINDSHIELD WIPER ARM AND BLADE REMOVAL/INSTALLATION.)